

EXP NO: 9
Date:22-9-26

Mini Project – Time Series-Based Stock Price Forecasting with Linear Regression

AIM:

To predict future stock prices using historical time-series data by implementing a **Linear Regression model** and evaluating its performance in forecasting stock price trends.

ALGORITHM:

Part 1: Hyperparameter Tuning using Grid Search

- ☐ **Data Collection:** Obtain historical stock price data such as Open, High, Low, Close, and Volume.
- ☐ **Data Preprocessing:** Handle missing values and convert date columns into the appropriate datetime format.
- ☐ **Sort Data:** Arrange the stock data in chronological order.
- ☐ **Feature Selection:** Select relevant features such as Open, High, Low, Close, and Volume.
- ☐ **Create Features:** Create lag features if required for stock price prediction.
- ☐ **Train-Test Split:** Divide the dataset into training and testing sets, such as 80% training and 20% testing data.
- ☐ **Model Training:** Train a Linear Regression model using the training data.
- ☐ **Prediction:** Use the trained model to predict stock prices for the test data.
- ☐ **Evaluation:** Evaluate the model using **MSE, RMSE, and R^2 score**.
- ☐ **Visualization:** Plot the actual and predicted stock prices to observe the forecasting performance.

CODE:

Index.html:

```
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5
6 <meta charset="UTF-8">
7
8 <meta name="viewport"
9   content="width=device-width, initial-scale=1.0">
10
11 <title>Stock Price Predictor</title>
12
13 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
14   rel="stylesheet">
15
16 <style>
17
18   body {
19
20     font-family: 'Arimo', sans-serif;
21
22     background-color: #f0f8ff;
23
24     padding: 50px;
25
26   }
27
28   .login {
29
30     max-width: 500px;
31
32     margin: auto;
33
34     background: white;
35
36     padding: 30px;
37
38     border-radius: 10px;
39
40     box-shadow: 0px 0px 10px 0px gray;
41
42   }
```

```

42     }
43
44     h1,
45     h2 {
46
47         color: #1e81b0;
48
49     }
50
51     .footer-dark {
52
53         text-align: center;
54
55         margin-top: 40px;
56
57         padding: 20px;
58
59         background-color: #1e81b0;
60
61         color: white;
62
63     }
64
65 </style>
66
67 </head>
68
69
70 <body style="background-color: #greenyellow; font-family: cursive;">
71
72     <div class="login"
73         style="background-color: #aqua;">
74
75         <h1 style="color: black;">
76
77             Stock Price Predictor
78
79         </h1>
80
81
82         <h2 style="color: black;">
83
84             Welcome, {{ username }}!
85
86         </h2>
87
88         <div class="mb-4 text-center">
89             

```

```

93
94
95
96     <form action="{{ url_for('dashboard') }}"
97         method="post">
98
99         <input type="number"
100             name="Open"
101             placeholder="Opening Price"
102             class="form-control mb-3"
103             required>
104
105
106         <input type="number"
107             name="High"
108             placeholder="Highest Price"
109             class="form-control mb-3"
110             required>
111
112
113         <input type="number"
114             name="Low"
115             placeholder="Lowest Price"
116             class="form-control mb-3"
117             required>
118
119
120         <input type="number"
121             name="Volume"
122             placeholder="Volume"
123             class="form-control mb-3"
124             required>

```

```

119
120     <input type="number"
121           name="Volume"
122           placeholder="Volume"
123           class="form-control mb-3"
124           required>
125
126
127     <button type="submit"
128           class="btn btn-primary w-100"
129           style="background-color: #28a745;">
130
131         Predict
132
133     </button>
134
135 </form>
136
137
138 <br>
139
140
141 <h4 class="text-success"
142     style="background-color: white; color: black;">
143
144     {{ prediction_text }}
145
146 </h4>
147
148 </div>
149
150
151 <div class="footer-dark">
152
153     <footer>
154
155         <div class="container">
156
157             <p class="copyright">
158
159                 Stock Price Predictor © 2025
160
161             </p>
162
163         </div>
164
165     </footer>
166
167 </div>

```

```

168
169
170     <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js">
171     </script>
172
173 </body>
174
175 </html>

```

Home.html:

```
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7
8      <title>Stock Price Prediction</title>
9
10     <link rel="stylesheet"
11         href="https://cdnjs.cloudflare.com/ajax/libs/animate.css/4.1.1/animate.min.css"/>
12
13     <link rel="stylesheet"
14         href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.2/css/all.min.css">
15
16     <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
17         rel="stylesheet">
18
19     <style>
20         .head {
21             color: #1e81b0;
22         }
23
24         .bt {
25             background-color: #1e81b0;
26         }
27
28         html {
29             scroll-behavior: smooth;
30         }
31     </style>
32 </head>
33
34 <body style="background-color: #greenyellow; font-family: cursive;">
35
36     <!-- Header Section -->
37     <div class="container">
38
39         <header class="d-flex flex-wrap justify-content-center py-3 mb-4 border-bottom">
40
41             <a href="/"
42                 class="d-flex align-items-center mb-3 mb-md-0 me-md-auto link-body-emphasis text-decoration-n
43
44                 
49
50                 <span class="fs-3 m-3 fw-bold head"
51                     style="color: red">
52
53                 <i class="fas fa-chart-line"></i>
```

```
53     <i class="fas fa-chart-line"></i>
54     Stock Price PredictionSss
55
56     </span>
57
58 </a>
59
60 <ul class="nav nav-pills">
61
62     <li class="nav-item">
63         <a href="{ url_for('home') }"
64             class="nav-link link-light"
65             style="background-color: red">
66             Home
67         </a>
68     </li>
69
```

```

70     <li class="nav-item">
71         <a href="#features"
72             class="nav-link"
73             style="color: red">
74             Features
75         </a>
76     </li>
77
78     <li class="nav-item">
79         <a href="#about"
80             class="nav-link"
81             style="color: red">
82             About
83         </a>
84     </li>
85
86     <li class="nav-item">
87         <a href="{{ url_for('register') }}"
88             class="nav-link"
89             style="color: red">
90             Register
91         </a>
92     </li>
93
94     <li class="nav-item">
95         <a href="{{ url_for('login') }}"
96             class="nav-link"
97             style="color: red">
98             Login
99         </a>
100     </li>
101
102     <li class="nav-item">
103         <a href="{{ url_for('login') }}"
104             class="btn btn-lg text-light fw-bold"
105             style="background-color: green">
106             Get Started
107         </a>
108     </li>
109 </ul>
110 </header>
111
112 </header>
113

```

```

1
2     </header>
3
4 </div>
5
6
7 <!-- Hero Section -->
8
9 <div class="container col-xxl-8">
10
11     <div class="row flex-lg-row-reverse align-items-center g-5 py-5">
12
13         <div class="col-10 col-sm-8 col-lg-6 animate animated animate zoomIn">
14
15             
21
22         </div>
23
24         <div class="col-lg-6 animate animated animate slideInLeft"
25             style="color: black;">
26
27             <h1 class="display-5 fw-bold lh-1 mb-3"
28                 style="color: red">
29
30             Stock Price Prediction
31

```

```

131
132
133
134 <div class="col-lg-6 animate animated animate slideInLeft"
135 style="color: black;">
136
137 <h1 class="display-5 fw-bold lh-1 mb-3"
138 style="color: red;">
139
140 Stock Price Prediction
141
142 </h1>
143
144 <p class="lead">
145
146 Stock Price Prediction estimates stock values
147 using machine learning models based on features
148 like opening price, highest price, lowest price,
149 volume, and market trends.
150
151 </p>
152
153 <div class="d-grid gap-2 d-md-flex justify-content-md-start">
154
155 <a href="/login">
156
157 <button type="button"
158 class="btn btn-lg px-4 me-md-2 text-light fw-bold"
159 style="background-color: green;">
160
161 Get Started
162
163 </button>
164
165 </a>
166
167 </div>
168
169 </div>
170
171 </div>
172
173 </div>
174
175
176 <!-- Features Section -->
177
178 <div class="container py-5" id="features">

```

```

178 <div class="container py-5" id="features">
179
180 <h2 class="text-center head animate animated animate fadeInDown"
181 style="color: red;">
182
183 Features
184
185 </h2>
186
187 <div class="row text-center">
188
189 <div class="col-md-4 animate animated animate zoomIn">
190
191 <h4 style="color: blueviolet;">
192
193 <i class="fas fa-chart-line"></i>
194 Accurate Predictions
195
196 </h4>
197
198 <p>
199 Our model uses advanced machine learning
200 algorithms to estimate stock prices with
201 high accuracy.
202 </p>
203
204 </div>
205
206
207 <div class="col-md-4 animate animated animate zoomIn">
208
209 <h4 style="color: blueviolet;">
210
211 <i class="fas fa-clock"></i>
212 Real-Time Data
213
214 </h4>

```

```

264         on key features like historical data, market
265         trends, and economic indicators.
266
267         Our goal is to provide reliable and insightful
268         predictions to help investors make informed
269         decisions.
270
271     </p>
272
273 </div>
274
275 <div class="col-md-6 animate animated animate fadeInRight">
276
277     
280
281 </div>
282
283 </div>
284
285 </div>
286
287
288 <!-- Footer Section -->
289
290 <div class="container">
291
292     <footer class="py-3 my-4">
293
294         <p class="text-center text-body-secondary">
295
296             &copy; 2025 Company, Inc
297
298         </p>
299
300     </footer>
301
302 </div>
303
304
305 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js">
306 </script>
307
308 </body>
309
310 </html>

```

Login.html:

```

1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5
6     <meta charset="UTF-8">
7
8     <meta name="viewport"
9         content="width=device-width, initial-scale=1.0">
10
11     <title>Login</title>
12
13     <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
14         rel="stylesheet">
15
16     <style>
17
18         body {
19
20             height: 100vh;
21
22             display: flex;
23
24             align-items: center;
25
26             justify-content: center;
27
28         }
29
30         .login-container {
31
32             border-radius: 10px;
33
34             overflow: hidden;
35
36         }
37
38     </style>
39

```

```

42
43 <body style="background-color: #greenyellow;">
44
45   <div class="container">
46
47     <div class="row login-container">
48
49       <!-- Left Side Image -->
50
51       <div class="col-md-6"
52         style="margin-top: 50px;">
53
54         
58
59       </div>
60
61
62       <!-- Right Side Form -->
63
64       <div class="col-md-6 p-5">
65
66         <div class="shadow-lg p-5 m-5 bordered rounded"
67           style="background-color: #aqua;">
68
69           <h2 class="text-center mb-4"
70             style="font-family: cursive;">
71             Login
72           </h2>
73
74           <form action="{ url_for('login') }}"
75             method="post">
76
77             <div class="mb-3">
78
79               <label for="username"
80                 class="form-label"
81                 style="color: #blue; font-family: cursive;">
82                 Username
83               </label>
84
85               <input type="text"
86                 class="form-control"
87                 name="username"
88               >
89
90
91
92

```

```

90       <input type="text"
91         class="form-control"
92         name="username"
93         id="username"
94         placeholder="Enter your username"
95         required>
96
97     </div>
98
99
100    <div class="mb-3">
101
102      <label for="password"
103        class="form-label"
104        style="color: #blue; font-family: cursive;">
105        Password
106      </label>
107
108      <input type="password"
109        class="form-control"
110        name="password"
111        id="password"
112        placeholder="Enter your password"
113        required>
114
115    </div>
116
117
118    <div class="d-grid">
119
120      <button type="submit"
121        class="btn"
122        style="background-color: #green; font-family: cursive;">
123        Login
124      </button>
125
126    </div>
127
128
129    <div class="text-center mt-3"
130      style="font-family: cursive;">
131
132      <p>
133
134
135
136
137

```

```

66 <p>
67
68     Don't have an account?
69
70     <a href="{{ url_for('register') }}"
71         class="text-primary">
72
73         Register
74     </a>
75
76 </p>
77
78 </div>
79
80 </form>
81
82 </div>
83
84 </div>
85
86 </div>
87
88 </div>
89
90 </div>
91
92 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js">
93 </script>
94
95 </body>
96
97 </html>

```

Register.html:

```

1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5
6      <meta charset="UTF-8">
7
8      <meta name="viewport"
9          content="width=device-width, initial-scale=1.0">
10
11      <title>Register</title>
12
13      <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
14          rel="stylesheet">
15
16      <style>
17
18          body {
19
20              height: 100vh;
21
22              display: flex;
23
24              align-items: center;
25
26              justify-content: center;
27
28          }
29
30          .register-container {
31
32              border-radius: 10px;
33
34              overflow: hidden;
35
36          }
37
38      </style>
39
40 </head>
41
42
43 <body style="background-color: #f0f0f0;">
44
45     <div class="container">
46
47         <div class="row register-container">
48
49             <!-- Left Side Image -->
50
51             <div class="col-md-6"
52                 style="margin-top: 150px;">
53
54                 
59
60
61     <!-- Right Side Form -->
62
63     <div class="col-md-6 p-5">
64
65         <div class="shadow-lg p-5 m-5 bordered rounded"
66             style="background-color: #aqua;">
67
68             <h2 class="text-center mb-4"
69                 style="color: black">
70                 Register
71             </h2>
72
73             <form action="{{ url_for('register') }}"
74                 method="post">
75
76                 <div class="mb-3">
77
78                     <label for="name"
79                         class="form-label"
80                         style="color: blue;">
81                     Name
82                     </label>
83
84                     <input type="text"
85                         class="form-control"
86                         name="username"
87                         id="name"
88                         placeholder="Enter your name"
89                         required>
90                 </div>
91
92                 <div class="mb-3">
93
94                     <label for="gender"
95                         class="form-label"

```

```

96                     style="color: blue;">
97                     Gender
98                     </label>
99
100                     <select class="form-control"
101                         name="gender"
102                         id="gender"
103                         required>
104
105                         <option value="">
106                             Select Gender
107                         </option>
108
109                         <option value="Male">
110                             Male
111                         </option>
112
113                         <option value="Female">
114                             Female
115                         </option>
116
117                         <option value="Other">
118                             Other
119                         </option>
120                     </select>
121
122                 </div>
123
124                 <div class="mb-3">
125
126                     <label for="age"
127                         class="form-label"
128                         style="color: blue;">
129                     Age
130                     </label>
131
132                     <input type="number"
133                         class="form-control"
134                         name="age"
135                         id="age"
136                         placeholder="Enter your age"
137                         required>
138                 </div>
139
140             </div>
141
142         </div>
143
144     </div>
145
146 </div>

```

```

177
178
179
180
181
182
183
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185
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194
195
196
197
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199
200
201
202
203
204
205
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212
213
214
215
216
217
218
219
220
221
222

```

```

<button type="submit"
  class="btn"
  style="background-color: #28a745;">
  Register
</button>
</div>

<div class="text-center mt-3">
  <p>
    Already have an account?
    <a href="{% url_for('login') %}"
      class="text-primary">
        Sign In
    </a>
  </p>
</div>
</form>
</div>
</div>
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js">
</script>
</body>
</html>

```

OUTPUT:


Stock Price Prediction

[Home](#)
[Features](#)
[About](#)
[Register](#)
[Login](#)
[Get Started](#)

Stock Price Prediction

Stock Price Prediction estimates stock values using machine learning models based on features like opening price, highest price, lowest price, volume, and market trends.

[Get Started](#)



Features



Accurate Predictions

Our model uses advanced machine learning algorithms to estimate stock prices with high accuracy.



Real-Time Data

Stay updated with real-time stock market trends and price changes.



User-Friendly

Easy-to-use interface for investors, traders, and analysts.



Login

Username

Password

Don't have an account? [Register](#)



Register

Name

Gender

Age

Password

Already have an account? [Sign In](#)

RESULT:

The **Stock Price Prediction** web application was successfully developed and implemented. The system allows users to register and log in, enter stock details such as **Opening Price, Highest Price, Lowest Price, and Volume**, and obtain a predicted stock price.

The application successfully displays the predicted stock price through the prediction page. The project demonstrates the implementation of a stock-price forecasting system with a user-friendly web interface.